



Final Exam

International Campus, KISH

MD Program, Biochemistry Metabolism

Semester, Sept 2017

Name and Surname:

Student ID Number:

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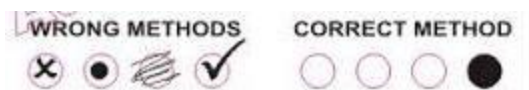
Duration of exam: 30 Minutes

MCQ Question

33

Directions:

- Each of the following questions contains one correct answer only.
- Please fill circle of the correct answer for the following questions on the answer sheet ONLY with **pencil**





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1. **What is the compound that comprises the majority of the nonprotein nitrogen fraction in serum?**
 - a. Urea
 - b. Uric acid
 - c. Amino acid
 - d. Bilirubin

2. **Failure of conversion of ammonia to urea is an indicator of which disease:**
 - a. Kidney failure
 - b. Congestive heart failure
 - c. End stage liver disease
 - d. Dehydration

3. donates its phosphate to ADP to yield ATP during muscle exertion
 - a. Creatine phosphate
 - b. Creatine kinase
 - c. Guanidoacetate
 - d. Arginine glycine amidinotransferase

4. **Tophus is a clinical sign of which condition?**
 - a. Hypouricemia
 - b. Hyperuricemia
 - c. Hyperbilirubinemia
 - d. Pre-hepatic jaundice

5. **The end product of pyrimidine synthesis is**
 - a. AMP
 - b. IMP
 - c. UMP
 - d. GMP

6. **Which of the following molecules below IS NOT a part of the synthesis of a purine ring**
 - a. Aspartate
 - b. CO₂
 - c. Glutamine
 - d. Lysine

7. **Ketogenic amino acids are called ketogenic because they degrade to**
 - a. Acetoacetyl coA and can form ketone bodies in the liver
 - b. Molecules that become oxaloacetate via TCA cycle
 - c. Oxaloacetate
 - d. Pyruvate



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- 8. Which of the following is a both ketogenic and glucogenic amino acid?**
- a. Ile, Trp
 - b. Leu, Phe
 - c. Tyr, Lys
 - d. Trp, Lys
- 9. All of the following conversions is performed by the enzyme phenylalanine hydroxylase EXCEPT?**
- a. Phe to Tyr
 - b. NADP^+ to NADPH
 - c. O_2 to H_2O
 - d. H4-biopterin to H2-biopterin
- 10. Which is NOT TRUE in the liver**
- a. Alanine converts to pyruvate
 - b. Alpha-ketoglutarate becomes glutamate
 - c. Pyruvate becomes converted to glucose
 - d. Glucose is used by live tissue to preproduce energy
- 11. The enzyme that converts glutamine to glutamate is called**
- a. Alanine aminotransferase
 - b. Glutamate dehydrogenase
 - c. Glutamine synthetase
 - d. Glutamainase
- 12. A common compound shared by urea and TCA cycle is**
- a. Fumarate
 - b. Succinyl CoA
 - c. Oxaloacetate
 - d. Alpha-ketoglutarate
- 13. Which of the following are the ketone bodies?**
- a. Acetyl co A and Propionyl co A
 - b. Lecithin and Lysolecithin
 - c. Acetoacetate and Betahydroxy butyrate
 - d. Pyruvate and lactate
- 14. Insulin promotes which of the following?**
- a. Transport of glucose in to cells
 - b. Conversion of Triacyl glycerol to diacylglycerol
 - c. Fatty acid oxidation
 - d. Increased blood glucose level



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- 15. A 2 year -old boy has been diagnosed with Zellweger syndrome, a disorder caused by malformation of peroxisomes. The oxidation of which of the following fatty acids is impaired in this defect?**
- Palmitic acid
 - Unsaturated fatty acids
 - Very long chain fatty acids**
 - Branched hail fatty acids
- 16. What is the precursor for cholesterol synthesis?**
- Acetyl CoA**
 - Propionyl CoA
 - Succinyl CoA
 - Acetoacetyl CoA
- 17. The acetyl CoA is produced in the mitochondria and must be transported into cytosol for synthesis of fatty acid. Which of the following is true regarding its transport?**
- Acetyl CoA is diffused from mitochondrial membrane
 - Acetyl CoA is transported by its specific transporter protein
 - Acetyl CoA is converted into pyruvate, enters into cytosol and acetyl CoA is regenerated
 - Acetyl CoA is converted into citrate, enters into cytosol and acetyl CoA is regenerated**
- 18. The dietary fats are transported as –**
- Micelles
 - Chylomicrons**
 - Fatty acid – Albumin complex
 - Liposomes
- 19. A fatty acid with 14 carbon atoms will undergo how many cycles of beta oxidation**
- 7
 - 4
 - 6**
 - 5
- 20. Which statement best describes the oxidation of odd chain fatty acids?**
- Additional specific enzymes are needed for the oxidative process
 - One carbon is removed in one cycle
 - End product is propionyl co A**
 - Hydroxy fatty acids are produced



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21. The key regulatory enzyme of cholesterol synthesis is-

- a. HMG- Co A synthase
- b. HMG Co A lyase
- c. HMG Co A reductase
- d. Mevalonate kinase

22. 50-year-old woman was brought to clinic with the diagnosis of bile ducts obstruction. The activity of what enzyme will prevail in her blood plasma?

- a. Alkaline phosphatase
- b. g-Glutamyl transpeptidase
- c. Aspartate aminotransferase
- d. Acidic phosphatase. E. Hexokinase

23. A patient presents high activity of LDH1, 2, aspartate aminotransferase, creatine phosphokinase. In what organ (organs) is the development of a pathological process the most probable?

- a. In the heart muscle (initial stage of myocardium infarction)
- b. In skeletal muscles (dystrophy, atrophy)
- c. In kidneys and adrenals
- d. In connective tissue E In liver and kidneys

24. What is enzyme is more specific for acute pancreatitis?

- a. AST
- b. Amylase
- c. GGT
- d. ALP

25-Where are the enzymes of Glyco lysis?

- a) Cytoplasm
- b) Mitochondria
- c) Out of the cells
- d) On the ribosome

26- How many pyruvate molecules are synthesized in glycolysis?

- a) 1
- b) 2
- c) 3
- d) 4

27- All the Following are anaerobic fermentation EXCEPT:

- a) Conversion of pyruvate to lactate
- b) Conversion of pyruvate to ethanol
- c) Conversion of pyruvate to acetyl CoA
- d) Synthesis of acetaldehyde



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28- Glucose can enter the muscle cell through which of the following transporters?

- a) GLUT 1
- b) GLUT 2
- c) GLUT 3
- d) **GLUT 4**

29- Which of the following enzymes is important in glycogen breakdown?

- a) Synthase
- b) Ligase
- c) Phosphatase
- d) **phosphorylase**

30- All the following enzymes are important in gluconeogenesis EXCEPT:

- a) fructose 1,6 biphosphatase
- b) glucose 6 phosphatase
- c) PEP carboxykinase
- d) Hexokinase

31- Reactions of TCA are located in...?

- a) cytoplasm
- b) **mitochondria**
- c) out of the cells
- d) on the ribosome

32- Which of the following reactions in TCA produces high energy electrons?

- a) isocitrate $\longrightarrow$ α - ketoglutarate
- b) citrate $\longrightarrow$ cis-aconitate
- c) fumarate $\longrightarrow$ malate
- d) cis- aconitate $\longrightarrow$ isocitrate

33- What is the importance of HMP SHUNT?

- a) Synthesis of glucose+NADH
- b) Synthesis of fructose+glucose
- c) **Synthesis of ribose+ NADPH**
- d) Synthesis of glycose+ FADH